

11/07/26

Short Answer
Test

CO1 - AT1

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Slot : D

Course Code : DSA0405

Course Name : Fundamentals of Data Science

1) What is Data science? Explain its primary objective.

* Data science is the process of collecting, analyzing and interpreting data using scientific methods, algorithms and technology. Its primary objective is to extract useful insights from data.

2) Why is Data science important in today's digital world? Mention any four reasons.

* Data science helps organizations make informed decisions based on data instead of assumptions. It improves business efficiency, predicts future trends, enhances customer experience and supports automation through AI.

3. List any four benefits of data science in business and society.

* Decision making

* Increases business profits

* Detects fraud

* Enhances healthcare services.

4)

Mention four real-world application of data science.

- * Healthcare for disease prediction
- * Banking for fraud detection
- * E-commerce for product recommendations
- * Transportation for traffic prediction.

5)

What are different forms of data? Briefly explain each.

* The main forms of data are structured, semi-structured and unstructured data. Structured data is organized in tables. Semi-structured data has partial organization like XML/JSON. Unstructured data includes images, videos and text.

6)

Differentiate between structured, semi-structured and unstructured data with suitable examples.

* Structured data is organized in rows and columns.

Ex: Student database, Excel sheet.

* Semi-structured data has tags or metadata but no fixed format.

Ex: XML, JSON.

* Unstructured data is no predefined format.

Ex: Images, videos, audio files.

7)

What is Big data? State its key characteristics.

* Big data refers to a extremely large and complex datasets that cannot be processed using traditional tools.

Its five characteristics are volume, velocity, variety, veracity and value.

8)

Explain the Big data Ecosystem: Name any four technologies used in it.

* The Big data ecosystem consist of tools used for collecting, storing, processing, analyzing and visualizing large amount of data. Some popular technologies are Hadoop, Spark, Hive and HBase.

9)

What are the major phase of the Data science process?

* The Data science process includes data collection, data cleaning, data analysis, model building, model evaluation and deployment.

10)

What is data Retrieval? Mention two common source from which data can be retrieved.

* Data Retrieval is the process of obtaining data from different storage system for analysis. Common sources include databases and web APIs.

11)

Why is data collection considered an important step in the Data science process?

* Data collection provides the raw information needed for analysis and model building. Accurate and relevant data improves the quality and reliability of results.

any four technologies

used for collecting
large amount of data,
and more.

process

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12.

What is data cleaning? List any four common data cleaning techniques.

* Data cleaning is the process of identifying and correcting errors in a dataset. Common techniques include removing duplicates, handling missing values, correcting inconsistent data and removing outliers.

13)

What is data integration? Why is it necessary in data science?

* Data integration is the process of combining data from multiple sources into a single dataset. It provides a complete view of information and improves the accuracy.

14) Explain data transformation with suitable examples.

* Data transformation is the process of converting data into a suitable format of analysis. Examples include converting text to numbers, normalizing values, and changing date formats.

15)

What is data analysis? Mention any four techniques used for data analysis.

* Data analysis is the process of examining data to

discover useful patterns and insights. Common techniques

include descriptive analysis, regression analysis, classification

and clustering.

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